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From Compliance to Leadership in Advancing Methane Accuracy Globally

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Abstract

The Oil and Gas industry bears a significant responsibility for climate change, given its substantial contribution to global emissions, especially methane, a potent GHG. As PETRONAS operates on a global scale, with emissions regulations such as carbon taxes implemented in certain countries, driving the advancement of its methane emissions management strategy. PETRONAS became a signatory of OGMP 2.0 in November 2022 and currently is on the verge of achieving the Gold Standard in methane reporting.

Accuracy enhancement in methane emission figures is achieved by progressing through successive stages (Level 3, Level 4 and Level 5), as outlined by the OGMP 2.0 framework. Level 3 represents the minimum approach for quantification across natural gas value chain facilities, relies on generic emission factors applied at the source level. To meet this requirement, an in-house developed tool has been established specifically for this purpose. Level 4 incorporates direct source-level measurements or combination of metering and engineering-based calculations. Level 5 builds on Level 4 by incorporating site-level direct measurements and conducting reconciliation between top-down and bottoms-up measurements.

The United Nations Environment Programme (UNEP) has recognized and accepted PETRONAS's proposed implementation plan, awarding it the distinguished Gold Standard Pathway for two consecutive years. This progress to date has been fueled by dedicated commitment and proactive initiatives, supported by an open-door policy that encourages collaboration with technical service providers and universities. At both Level 4 and Level 5, various technologies were evaluated, tested and implemented such as portable QOGI, satellite, drone, etc. The real challenge is methane emissions are highly dynamic, ranging from trace levels in parts per million to over 100 grams per hour, with thousands of potential leaks contributing to this variability. No single technology or device may be sufficient to fulfil all requirements necessary to achieve Level 4 and 5. A combination of devices, such as QOGI and optical flare efficiency cameras, has been employed to meet the requirements, for example, to achieve Level 4 ground measurement coverage. The offshore environment and cloud cover are part of the challenges for Level 5 measurement. Devices that are compatible with offshore environments have been identified and used for site-wide methane plume detection and quantification, together with optimized drone flight planning to capture flare height and full plume dispersion.

PETRONAS is continuously advancing and striving for improvement and enhancement in methane emissions management. Benefiting from personnel upskilling and hands-on experience with technology testing and deployment, PETRONAS is maturing its collaborative efforts with other Southeast Asian countries to develop the region's first methane emissions technology evaluation center (METEC). METEC is targeted to serve as the key enabler for achieving high-accuracy methane data. Accurate methane data can enhance the efficiency and speed of emission reduction initiatives.

Introduction

Since its establishment in 2022, the Methane Taskforce at PETRONAS has served as a strategic platform to oversee and strengthen methane emissions management across the organisation. Comprising focal representatives from the Corporate Sustainability Office (CSO), Upstream and Gas Business Units, Group Health, Safety, Security and Environment (GHSSE) and Malaysia Petroleum Management (MPM) HSE (for the management of Petroleum Arrangement Contractors (PACs)), the taskforce is led by the Custodian Process from Project Technology & HSSE. Its primary objective is to ensure effective coordination and delivery of key initiatives aligned with PETRONAS' commitment to the OGMP 2.0 framework. Central to this effort is the methane emissions management framework, which encompasses four key pillars: **Measurement and Quantification**, involving the detection and quantification of emissions from individual sources up to the enterprise level using both direct and indirect techniques; **Verification**, ensuring the credibility and transparency of emission data through validation of source data, quantification methods, and work processes; **Reporting**, which entails publishing complete methane emission data to meet stakeholder requirements; and **Reduction**, focused on eliminating or minimising emissions from identified sources.

Understanding and quantifying methane emissions is fundamental in managing greenhouse gas emissions—particularly in the oil and gas sector. To drive meaningful reductions under decarbonization initiative, operators must first know where and how methane is being released. The UNEP provides guidance on categorizing methane emissions across the natural gas value chain, helping industry players prioritize mitigation efforts.

Typical methane emission sources include:

1. Flaring - Incomplete combustion during emergency or routine flaring activities can result in unburned methane emissions.
2. Fugitive Component and Equipment Leaks - Unintended methane leaks from valves, flanges, connectors, seals, and piping, typically detected through leak detection and repair (LDAR) programs.
3. Loss of Primary Containment - An unplanned or uncontrolled release of methane from primary containment.
4. Stationary Combustion - Methane slip from engines, turbines, and heaters due to incomplete combustion.
5. Gas-Driven Pneumatic Devices - Devices using pressurized natural gas for actuation often vent methane directly to the atmosphere.
6. Compressor Seals - Leakage from wet and dry seals on centrifugal and reciprocating compressors.
7. Glycol Dehydrators - Methane is released during the regeneration of glycol used to remove water vapor from natural gas streams, particularly in older or non-optimized units.
8. Well-Related Venting - Includes planned or unplanned venting during well completions, interventions, or blowdowns.
9. Tanks - Emissions from storage tanks due to working losses, breathing losses, or overpressure events.
10. Main Vent - Large, centralized vent systems used in certain process facilities or during depressurization.
11. Other Vents (Local Vents) - Smaller, localized vents from process skids, manifolds, or safety systems.

Together, these sources account for a significant portion of total methane emissions in upstream and gas business operations. Accurate identification, measurement, and reporting of these sources is a prerequisite for designing effective mitigation strategies—and progressing toward global climate targets.

Figure 1 shows the five OGMP 2.0 reporting levels. Levels 1 and 2 represent high-level estimation approaches, and the associated uncertainties in actual methane emissions remain significant at these tiers. The three highest reporting levels—Level 3, Level 4, and Level 5—involve measurements at individual emission sources, reflecting a progressive increase in the granularity and comprehensiveness of methane emissions data. To meet the minimum reporting requirements under the OGMP 2.0 framework, member companies are expected to progress to Level 3 reporting, which provides source-level, measurement-informed estimates.

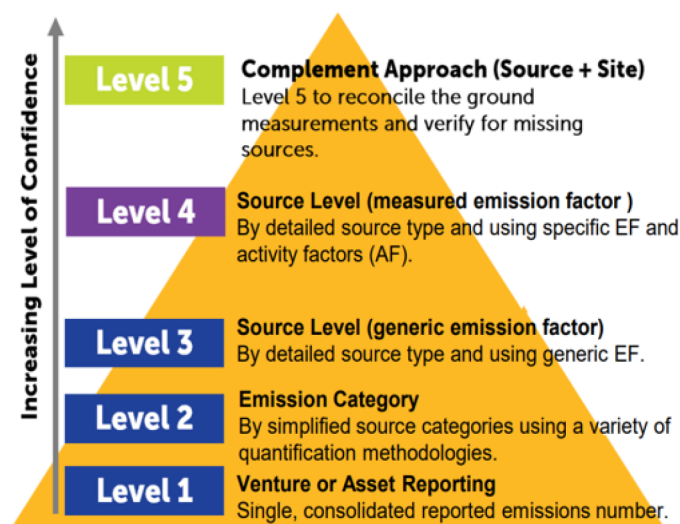


Figure 1—The five OGMP 2.0 reporting levels

In addition, achieving Level 3 is essential for conducting a robust materiality assessment, enabling the identification of material assets. Under OGMP 2.0 Framework, materiality refers to the significance of emissions for assets in terms of total emissions. All assets that account for 95% of total emissions for a given operator are considered material while the subset of assets that account for less than 5% of emissions can be considered as immaterial because they have a significantly small contribution to total emissions from a given operator. (OGMP, 2020). Emissions from immaterial assets must still be reported; however, there is no requirement to advance these assets to Level 4 or 5 reporting. The materiality assessment is a critical milestone on the pathway toward achieving the OGMP 2.0 Gold Standard, which emphasizes transparency, accuracy, and completeness in methane emissions reporting.

Member companies are expected to meet the Gold Standard reporting requirements within a timeline of three years for operated assets. For non-operated joint ventures (NOJVs) assets such as those under petroleum arrangement contracts (PACs), a longer timeframe of up to five years is allowed to meet the reporting requirements.

Methodology

The focus of this paper lies in the three highest levels - Levels 3, 4, and 5 - which provide source-level granularity by attributing emissions to specific source types such as gas-driven pneumatic devices and compressor seals, as outlined in the Introduction section.

The highlights on methodologies, tools, and measurement devices — each tailored to specific source types and site conditions — are drawn from PETRONAS' hands-on experience in progressing through the

OGMP 2.0 Gold Standard journey. These practical insights reflect the challenges and solutions encountered in both operated and NOJVs assets, spanning diverse environments across the natural gas value chain. The approach emphasizes fit-for-purpose deployment, balancing technical feasibility, safety, and cost-efficiency, while ensuring alignment with OGMP's rigorous reporting levels.

OGMP 2.0 Level 3

Level 3 marks the beginning of a detailed and comprehensive inventory of all potential methane emission sources which were developed based on P&ID (Piping and Instrumentation Diagram) reviews. Each emission source or component is linked to its respective process stream and associated composition. Emission estimates are performed at the source level using standard engineering formulas, typically applying:

$$\text{Emission} = \text{ComponentCount} \times \text{GenericEmissionFactor}$$

iCON Methane Emission Quantification Tool
(OPU: Test1)

Reported OPU: **Test1**
Industry Segment: **Production**

Tool Content and Navigation

(i) OPU Information

(ii) Methane Emission Sources

Unintended Release

- (1) Fugitive Emission**
 - 1.1 Quantification Method Selection: [Level 3\(a\) \(Screened Value Correlation\)](#)
 - 1.2 Detection and Concentration Measurement Devices: [Detector: OGI](#)
[Measurement Device: LEL/FID](#)
 - 1.3 [H&MB and Leak Inventory per Stream](#)
 - 1.4 [LDAR/GLOC Record](#)
- (2) Loss of Primary Containment (LOPC Tier 3)**
 - 2.1 [LOPC Record](#)
- (3) Main Vent Header**
 - 3.1 [Main Vent Header Record](#)
- (4) Stationary Combustion**
 - 4.1 [Stationary Combustion Record](#) [Fuel Property Database](#)
- (5) Main Flare Header**
 - 5.1 [Main Flare Header Record](#) [Flare Mixture Database](#)
- (6) Natural Gas Driven Pneumatic Equipment**
 - 6.1 [Natural Gas Driven Pneumatic Equipment](#)
- (7) Compressor Seals**
 - 7.1 [Compressor Seals Record](#)
- (8) Glycol Dehydrators**
 - 8.1 [Glycol Dehydrators Record](#)
- (9) Tank Breather Valves**
 - 9.1 [Tank Breather Valves Record](#)
- (10) Well Related Ventings**
 - 10.1 [Well Liquid Unloading Template](#)
 - 10.2 [Well Casinghead Venting Template](#)
 - 10.3 [Well Hydraulic Fracturing Template](#)
 - 10.4 [Well Testing Template](#)
- (11) Venting - Others**
 - 11.1 [Venting - Others Record](#)

(iii) Production Record

Figure 2—Landing page of MCON tool

PETRONAS developed its own Level 3 methane emissions calculation tool known as MCON for consistency and standardization across its operations. The tool provides flexibility in reporting based on user needs or alignment with the existing reporting templates, allowing for use as a single file per location or per asset, depending on the defined emissions boundary for reporting.

Figure 3 illustrates the process workflow for Level 3 methane quantification, incorporating the use of the MCON tool. The workflow begins with documents and data gathering, followed by systematic identification of emission sources from P&IDs, mapping of each source to its associated stream composition for methane content input and finally auto-calculation using embedded emission factors and consolidation of results within the tool.

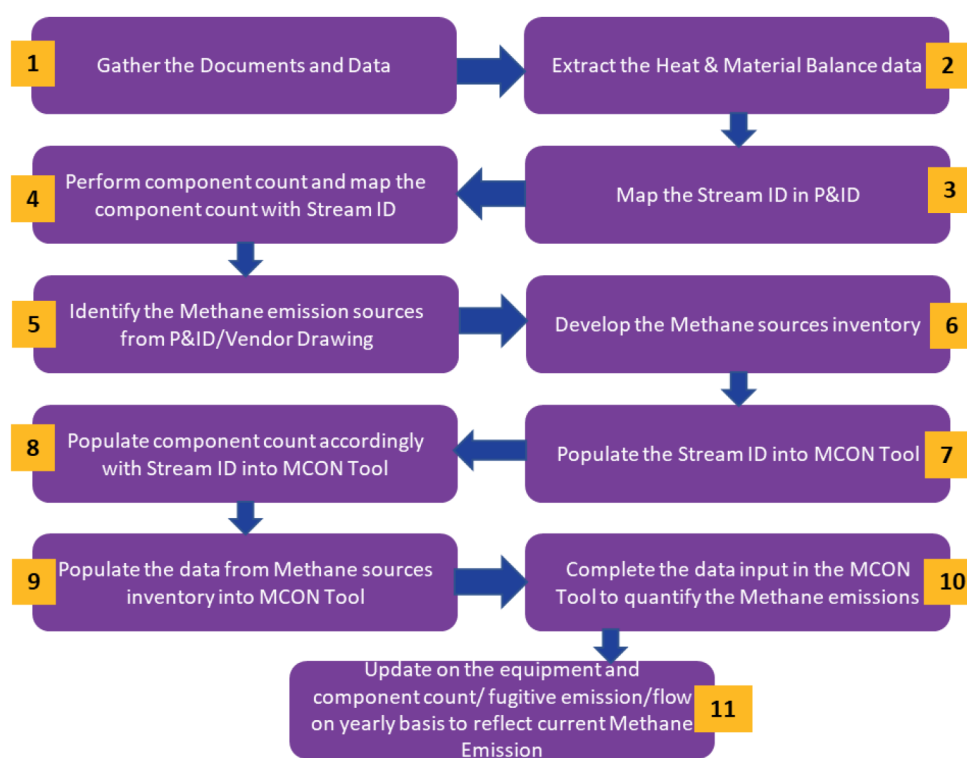


Figure 3—Process workflow for OGMP 2.0 Level 3 methane quantification

Once the tool is customized for a specific asset, any future updates such as incorporating fugitive leakers identified through the Leak Detection and Repair (LDAR) program—can be efficiently managed by updating Step 11 only. Step 11 includes as follows:

- a. Equipment and components count (addition or reduction in quantity)
- b. Newly identified leak sources under fugitive leaks and LOPC categories
- c. Process flow (depending on production profile)
- d. Operating hours
- e. Flaring /Venting/Stationary combustion flow and compositions

OGMP 2.0 Level 4

Level 4 represents the transition from estimation to measurement-based approaches by incorporating actual site data, resulting in more representative and accurate emission figures. At this stage, a combination of technologies and techniques is often deployed, tailored to the specific emission source and physical constraints of the site, such as hard-to-reach areas, elevated equipment, or hazardous zones.

Fugitive emissions, in particular, are typically addressed through Leak Detection and Repair (LDAR) programs, where leaks are identified, quantified, and subsequently rectified as part of routine maintenance and compliance efforts.

PETRONAS' Level 4 measurement journey began by leveraging existing process safety procedures through LDAR surveys using FLIR cameras in combination with LEL meters. Building on this foundation, PETRONAS expanded its efforts by testing Quantification Optical Gas Imaging (QOGI) portable devices and FLIR camera paired with quantification tablet. Following successful trials, PETRONAS invested in QOGI for its Upstream assets and conducted a series of training sessions to equip Inspection focals—who serve as the primary users of the devices during LDAR or Gas Loss of Containment (GLOC) surveys. These capabilities have since been extended beyond domestic operations, benefiting PETRONAS' international assets in Indonesia and Turkmenistan. The journey continues, as PETRONAS remains actively engaged in

testing and evaluating new generations of QOGI technology to ensure its practices remain at the forefront of methane emissions management.

Over the past years, this journey has been further strengthened through the development and application of a range of complementary measurement and estimation techniques across PETRONAS' operations enabling a comprehensive approach to methane quantification. This collective methodology integrates direct measurement such as flowmeters, flare meters, clamp-on sensors, and advanced optical gas imaging—with advanced engineering estimation methods, including gas balance calculations, flaring destruction efficiency and process simulation models calibrated with site-specific operational data. By leveraging both approaches, PETRONAS ensures that every significant methane source, whether continuous or intermittent, is accurately characterized.

The synergy between these methods not only provides a robust and verifiable pathway toward OGMP 2.0 Level 4 reporting but also reinforces confidence in the accuracy and integrity of reported methane figures. Within PETRONAS' operations, the primary contributors to methane emissions include main venting, flaring, gas-driven pneumatic devices, compressor seal vents, and tank breather valve venting. Each of these sources has been assessed, and targeted measurement or estimation methodologies have been deployed to ensure accurate, verifiable, and high-integrity reporting.

For main venting, direct quantification can be achieved by installing flowmeters to measure the vent gas flow combined with methane content analysis. Where direct measurement is challenging, an alternative approach integrates a gas balance with process simulation to estimate vented volumes.

For flaring, accurate reporting requires both direct flow measurement using a flare flowmeter and engineering calculations to determine the actual destruction efficiency of the flare system, ensuring reported values reflect true methane releases.

For gas-driven pneumatic devices, bleed rates can be directly measured by installing clamp-on meters at the instrument air or gas header. High-sensitivity True RMS meters, capable of detecting minimal AC or DC leakage currents, enable precise detection of low-level emissions from these devices.

Compressor seal vents are quantified using Quantitative Optical Gas Imaging (QOGI), which allows direct visualization and measurement of methane at the source, enabling accurate assessment of seal-related emissions.

Lastly, tank breather valve venting is best addressed through an indirect approach, using process simulation informed by site-specific operational data such as ambient temperature, tank throughput, liquid level, and tank geometry. This enables prediction of methane emissions for different tank roof configurations, even where direct measurement is not feasible.

The table below summarizes the key sources, methodologies, and technologies required to achieve OGMP 2.0 Level 4 source-level reporting for quick reference:

Table 1—Summary of key sources, methodologies and technologies required to achieve OGMP 2.0 Level 4 reporting

Source	Methodology / Approach	Technology
1. Main venting	- Install flowmeter to measure gas flow - Measure methane content in the stream - Alternatively, apply gas balance and process simulation model to estimate vented volume	- Direct measurement for flow - Indirect calculation via combination of gas balance and simulation
2. Flaring	- Install flare flowmeter to measure flow - Apply engineering calculation to determine actual flaring destruction efficiency	- Direct flow measurement - Indirect calculation of destruction efficiency
3. Gas-driven Pneumatic Devices	Install clamp-on meter at instrument header to measure bleed rate	Direct measurement using high-sensitivity True RMS meters capable of detecting low AC or DC leakage currents down to 1 mA

Source	Methodology / Approach	Technology
4. Compressor Seal Vents	Identify and measure emissions using Quantitative Optical Gas Imaging (QOGI)	Direct source measurement with gas visualization technology
5. Tank Breather Valves Vents	Quantify emissions via process simulation based on site data (ambient temperature, tank throughput, level indicator) and tank geometry	Indirect measurement via simulation predicting methane release for different tank roof types

OGMP 2.0 Level 5

Level 5 is the highest tier of the OGMP 2.0 framework, focusing on comprehensive site-level measurement that is further complemented by ground-based assessments to identify and verify any potential missing sources. Demonstrating PETRONAS' proactive commitment, PETRONAS began exploring top-down measurement technologies well before signing up for OGMP 2.0, in line with adherence to the Methane Guiding Principles at that time.

Back in 2020, PETRONAS explored the feasibility of using satellite-based measurements to achieve complete coverage across all site locations. However, at that time, satellite technology proved less compatible with offshore facilities due to interference from water surface reflections. This early trial highlighted the need for a diversified measurement approach. Over the following years, PETRONAS tested multiple devices and technologies, including fixed QOGI camera for both scenarios; onshore and offshore facilities and continuous monitoring of flare destruction efficiency using hyperspectral camera, initiatives that were supported through collaboration with a technical service provider, providing PETRONAS with valuable experience and opportunities to enhance its methane detection capabilities.

In parallel, PETRONAS embarked on a structured journey toward achieving OGMP 2.0 Level 5, which represents the Gold Standard in methane emissions reporting. This effort was underpinned by numerous stakeholder engagements, internal workshops, and lessons learned from other OGMP members that had successfully achieved Gold Standard reporting. Most recently in June 2025, PETRONAS hosted the ASEAN Energy Sector Methane Leadership Program (MLP) Technical Workshop on Methane Emissions Management in partnership with the Japan Organization for Metals and Energy Security (JOGMEC) and other MLP partners. The workshop provided a collaborative platform for regional energy companies to exchange experiences, share lessons learned, and explore pathways to strengthen methane emissions measurement, monitoring, and reporting, while also highlighting the latest advancements in methane management technologies.

Today, at Level 5, PETRONAS employs advanced remote sensing technologies such as drones equipped with high-precision methane sensors and aircraft-mounted Gas Mapping LiDAR (GML) systems. These tools enable full site-wide methane plume detection and quantification, capturing both persistent and intermittent emission events. Drone flight paths are strategically planned to account for variables such as flare stack height and to ensure complete coverage of plume dispersion patterns.

This integrated aerial and ground-based measurement strategy not only enhances detection accuracy but also provides an independent verification layer for reported emissions, solidifying the credibility of PETRONAS' methane inventory at the highest reporting standard.

Figure 4 shows the OGMP 2.0 Level 4 and Level 5 measurement technologies evaluation through effective collaboration with technology service providers. PETRONAS is also working in collaboration with a leading university to develop country-specific methane emission factors that reflect local operating conditions, equipment profiles, and environmental variables. This initiative aims to replace the reliance on generic or global default factors, ensuring that emission calculations are more representative and defensible for national reporting and international benchmarking.

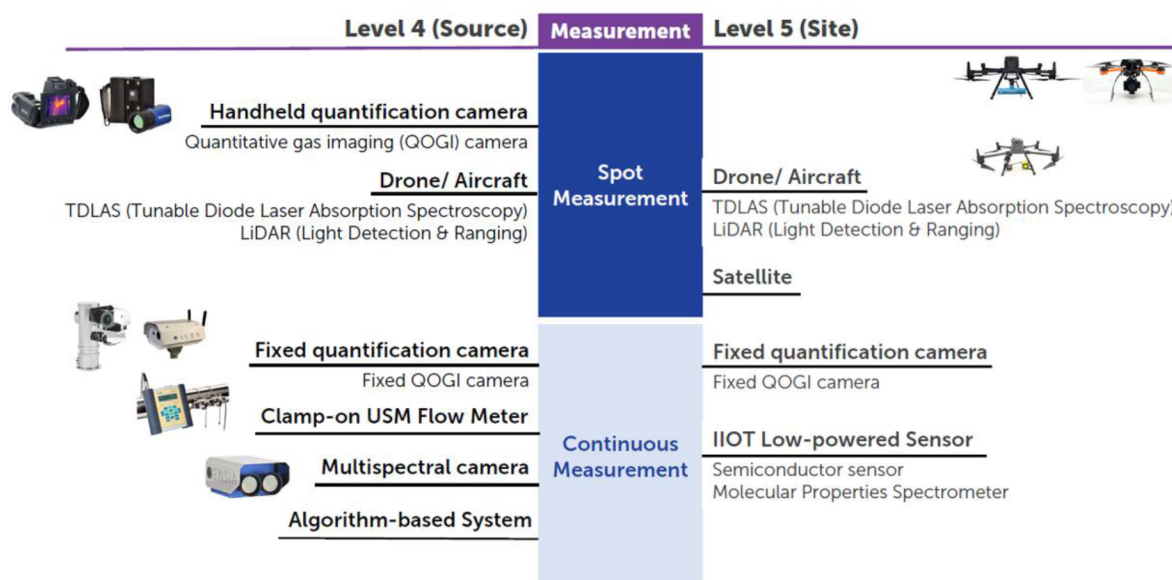


Figure 4—Technologies evaluation by PETRONAS for OGMP 2.0 Level 4 and Level 5 measurement

PETRONAS is also committed to continuing its testing and validation of emerging methane detection technologies, recognizing that rapid advancements in sensors, analytics, and remote monitoring can significantly enhance measurement accuracy. As part of this long-term vision, plans are underway to establish the Southeast Asia Methane Technology Left, a dedicated hub for research, technology trials, capacity building, and knowledge sharing. This left will serve not only as proving ground for innovative solutions but also as a regional platform to drive collaborative action on methane emissions reduction, positioning PETRONAS at the forefront of both national and regional climate leadership.

Case Study Examples for OGMP 2.0 Level 5 Reconciliation

Site level measurement was performed using drone mounted tunable diode laser absorption spectrometer (TDLAS) sensor to quantify the methane release across one of the Upstream fields. The survey was conducted over two days, covering three platforms where two runs were conducted for each platform. Collectively, the measured platforms represented the emissions and operational variability of the entire asset.

Reconciliation Results

Figure 5 shows a sample of case study finding from Level 5 reconciliation exercise at one of PETRONAS Upstream assets. The discrepancies between source-level and site-level measurements were further investigated to identify their underlying causes and to establish a clear justification for the observed differences.

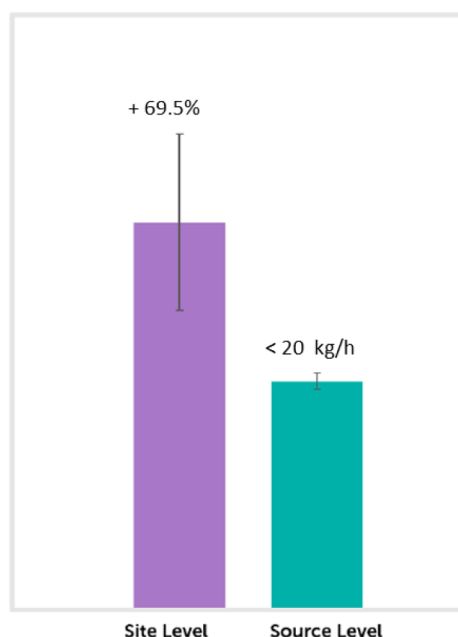


Figure 5—Sample case study of Level 5 reconciliation at one Upstream field

During the LDAR exercise, no fugitive leaks were detected using the QOGI camera. However, a leakage point was identified below the cooler deck during the drone inspection. This leak was subsequently confirmed by the QOGI camera and was promptly rectified upon confirmation.

In another wellhead platform equipped with natural gas pneumatic control system, drone measurement recorded significant methane emissions during the survey. Operational records show slickline activity was ongoing during the site measurement and it is suspected that this activity was the source of methane emissions. Potential emission sources are from the installation of the slickline or purge gas release during the well opening.

Another observation was that the drone survey measurement recorded zero detectable emissions, in contrast to the Level 4 source- level measurements, which indicated routine flaring activity. The drone flew at altitudes of 47 to 82 meters, while the flare stack height is approximately 54 meters and it was suspected that the plume may be above 82 meters. In addition, the wind speed was recorded at approximately 2.5 m/s indicating below the optimal range windspeed for drone measurement which is around 3 m/s.

Improvement Opportunities

Site Level Real Time Measurement Capability. Drone measurement should provide real time measurement results during the survey to enable immediate identification and confirmation of emission sources.

Refinement of Measurement Duration. Drone Limited measurement runs, particularly at platforms with significant emissions, have been observed to impact data reliability. Enhanced planning is necessary to ensure enough measurement frequency and comprehensive data collection, considering operational constraints such as weather and shutdown schedules.

Measurement Timing. Avoid scheduling measurements during process upsets, platform shutdowns or any activities to maintain data accuracy and data representativeness.

Integration of Anemometer Technology. The use of drone-mounted anemometers was identified as a critical factor for reducing measurement uncertainties. Future procedures should prioritize the integration of this technology to provide real-time wind data and improve data accuracy compared to ground-based anemometers.

Conclusion

Journey of PETRONAS in advancing methane emissions measurement reflects a deliberate shift from compliance-driven reporting towards proactive and efficient methane emissions management. Through the continuous testing and deployment of diverse methane measurement technologies, ranging from direct source-level quantification tools to advanced remote sensing systems, PETRONAS has progressively strengthened its ability to detect, quantify, and reconcile methane emissions with high accuracy and integrity.

This sustained commitment is not only about meeting OGMP 2.0 Gold Standard reporting requirements, but also about building a resilient measurement ecosystem that can adapt to evolving industry standards and emerging technologies. The lessons learned from field trials, technology evaluations, and reconciliation exercises have reinforced the importance of combining direct measurements with engineering estimations, integrating aerial, ground, and simulation-based methods, and fostering cross-disciplinary like technical service providers and university collaborations.

Looking ahead, the establishment of the Southeast Asia Methane Emissions Technology Evaluation Left will serve as a pivotal platform for research, innovation, and capability building in the region. By providing a dedicated hub for technology testing, data analysis, and knowledge sharing, the Left will accelerate the adoption of fit-for-purpose solutions and promote harmonized methane measurement practices across Southeast Asia.

Ultimately, these initiatives position PETRONAS not only as a leader in accurate and transparent methane reporting but also as a catalyst for regional collaboration in reducing methane emissions result in contributing meaningfully to both national commitments and global climate goals.

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